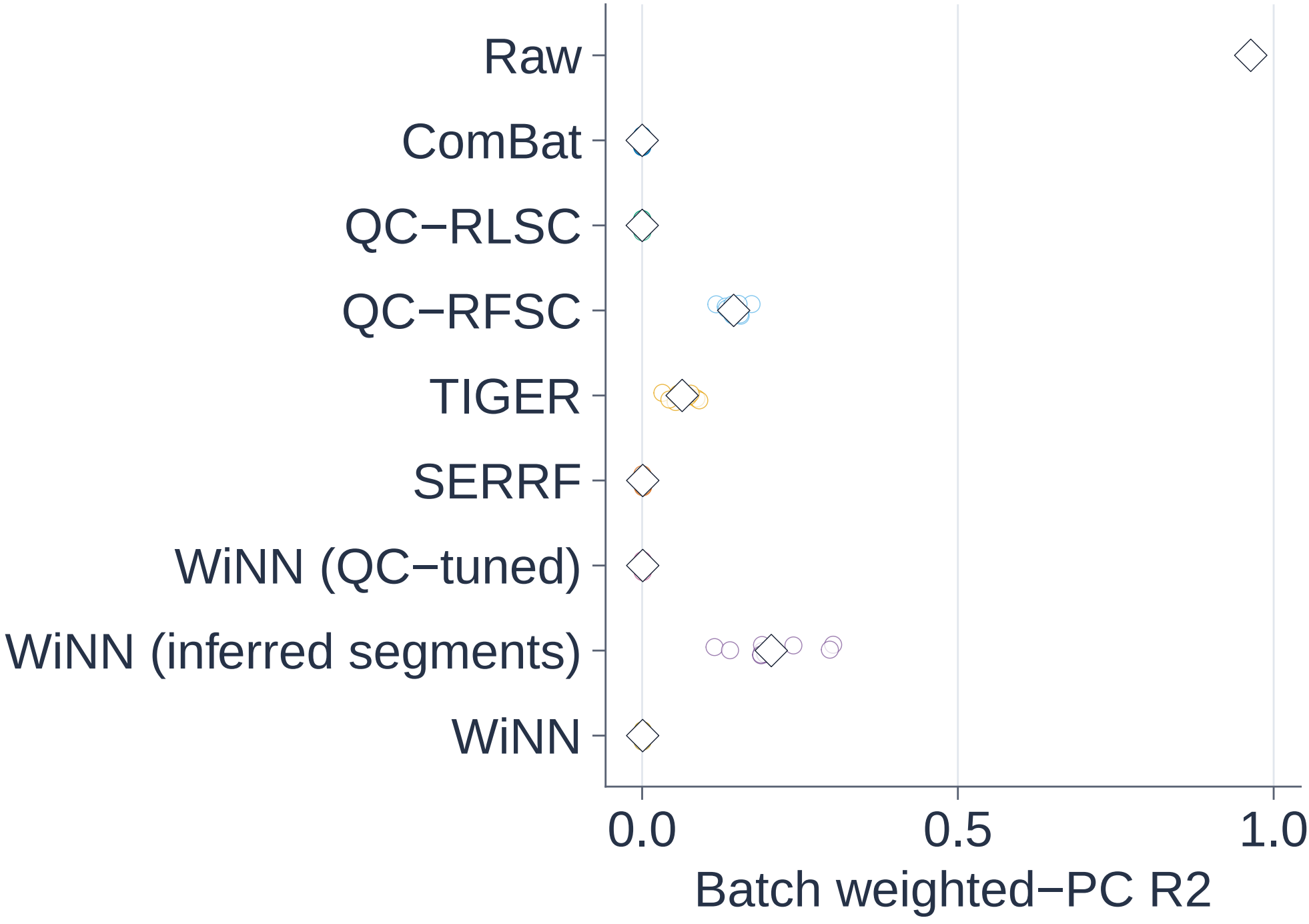C Residual batch structure